



1. **DESCRIPTION:** Participants will be assessed on their understanding of the anatomy and physiology for the **respiratory, digestive and immune systems of the human body.**

A TEAM OF UP TO: 2

APPROXIMATE TIME: 50 minutes

CALCULATOR: Class II

2. **EVENT PARAMETERS:**

- a. Each team may bring writing utensils.
 - b. Each team may bring two Class II calculators.
 - c. Each team may bring one (1) 8.5" x 11" sheet of paper that may contain information on both sides in any form and from any source. The sheet of paper may be laminated or placed in a sheet protector to increase durability. Affixed labels, as well as multiple sheets of paper (whether in a single sheet protector or not) are prohibited.
3. **THE COMPETITION:** This Event may be administered as a written test or as a series of lab-practical stations which can include but are not limited to experiments, scientific apparatus, models, illustrations, specimens, data collection and analysis, and problems for students to solve.

a. **RESPIRATORY SYSTEM:**

- i. Anatomy and functions of the respiratory system
- ii. Mechanisms and measures of pulmonary ventilation
- iii. Patterns of breathing
- iv. Ability to read a spirogram as related to pulmonary ventilation
- v. Gas exchange and O₂ transport including oxygen dissociation curves
- vi. Effects of exercise and high altitude on the respiratory system
- vii. Understand diseases/disorders: COPD, asthma, emphysema, pneumonia, sleep apnea and cystic fibrosis

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- viii. *Additional diseases/disorders: tuberculosis, pulmonary edema, pleurisy, small cell and non-small cell lung cancer, pulmonary fibrosis, pneumoconiosis (silicosis, coal worker lung, asbestosis)*
- ix. *Blood chemistry and respiratory rhythm, regulation and control of the respiratory system*
- x. *Treatments and/or preventions (drugs, surgery, etc.) for ALL diseases/disorders listed above*

b. **DIGESTIVE SYSTEM:**

- i. Anatomy and functions of the digestive system
- ii. Basic anatomy of the component parts of the alimentary canal and accessory organs of digestion
- iii. Anatomy of the four layers of the wall of the alimentary canal
- iv. Comparison of the lining of the esophagus, stomach, small intestine and large intestine
- v. Compare and contrast mechanical and chemical digestion
- vi. Physiology of chemical digestion of proteins, fats and carbohydrates
- vii. Effects of exercise and obesity on the digestive system
- viii. The diseases on each level from the cell to the whole person as listed: stomach & duodenal ulcers, cancers of the digestive system, diarrhea, lactose intolerance, hepatitis, appendicitis

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- ix. *Additional diseases: diverticular disease, GERD, peptic ulcer disease, ulcerative colitis, Crohn's Disease and celiac disease*
- x. *The function of the liver and pancreas in the digestive system, including Kupffer cell function*
- xi. *Treatments and/or prevention (drugs, surgery, etc.) for ALL diseases listed above*



c. IMMUNE SYSTEM:

- i. Anatomy and functions of the immune system (bone marrow, thymus, spleen, lymph nodes, skin)
- ii. Anatomy and physiology of innate immune system (eg. anatomical barriers, complement system, Toll-like receptors, inflammation, and innate immune cells)
- iii. Anatomy and physiology of adaptive immune system
- iv. Physiology of the immune response and allergic reactions
- v. Disorders: acquired and genetic immunodeficiencies (HIV/AIDS, common variable immunodeficiency, severe combined immunodeficiency), autoimmune diseases (multiple sclerosis, rheumatoid arthritis, Hashimoto thyroiditis, Graves' Disease), and type 1-4 hypersensitivities (anaphylaxis, urticaria, myasthenia gravis, Arthus reaction, serum sickness, contact dermatitis)

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- vi. *Types of Organ Transplants and Prevention of Rejection (allograft and autograft)*
- vii. *Cancer Immunotherapies (Antibody therapies, CAR-T cell therapy)*
- viii. *Additional disorders: systemic lupus erythematosus, dermatomyositis, scleroderma, psoriasis and psoriatic arthritis*
- ix. *Treatments and/or prevention (drugs, surgery, etc.) for ALL disorders listed above*

4. **SCORING:**

- a. High Score wins.
- b. Preselected questions will be used to break ties.

B